

Experiment 1: Bubble Sort – Exam Result Slip Correction

Aim

To implement the **Optimized Bubble Sort** algorithm in Python to sort student roll numbers using an **early-exit (swap flag)** technique.

Problem Statement

A teacher has a nearly sorted list of student roll numbers where only a few numbers are out of place. Use **Optimized Bubble Sort** to sort the list efficiently and display the number of passes required.

Algorithm

Step 1: Start.

Step 2: Read the number of roll numbers.

Step 3: Read the roll numbers into an array.

Step 4: Initialize passes = 0.

Step 5: Repeat Bubble Sort:

- Compare adjacent elements.
- Swap if they are in the wrong order.
- Set a swap flag if any swap occurs.

Step 6: If no swaps occur in a pass, stop the algorithm.

Step 7: Display the sorted roll numbers and the number of passes.

Step 8: Stop.

Source Code (Python)

```
main.py    Share  Run

1 # Optimized Bubble Sort
2
3 n = int(input("Enter number of roll numbers: "))
4 arr = []
5
6 print("Enter roll numbers:")
7 for i in range(n):
8     arr.append(int(input()))
9
10 passes = 0
11
12 for i in range(n - 1):
13     swapped = False
14     passes += 1
15
16     for j in range(n - i - 1):
17         if arr[j] > arr[j + 1]:
18             arr[j], arr[j + 1] = arr[j + 1], arr[j]
19             swapped = True
20
21     if not swapped:
22         break
23
24 print("Sorted Roll Numbers:", arr)
25 print("Number of Passes:", passes)
```

Output

```
Output

Enter number of roll numbers: 8
Enter roll numbers:
101
102
104
103
105
107
106
108
Sorted Roll Numbers: [101, 102, 103, 104, 105, 106, 107, 108]
Number of Passes: 2

=== Code Execution Successful ===
```

Complexity Analysis

Aspect	Complexity
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Best Case	$O(n)$
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Worst Case	$O(n^2)$
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Space Complexity $O(1)$

Result

The Optimized Bubble Sort algorithm was successfully implemented. The program sorted the roll numbers correctly and terminated early when no swaps were required.